Meet Mevada

+1 (630)-550-9574 | mmevad2@uic.edu
linkedin.com/in/meet-mevada-54bb87179/ | github.com/Breakpoint23

Summary

A keen learner with a **publication** in **machine/deep learning** and proven abilities in **control systems and automation** and proficient in various **programming languages** like **Python, MATLAB, Simulink and c++**. Proactive Engineer with experience in evolving innovative solutions for uncharted problems along with troubleshooting complex mechanical, electronic systems and **software programs**. Seeking a work environment where I can attain knowledge about state-of-the-art technologies in relevant fields and implement them while being innovative.

Skills

Programming Languages: Python, MATLAB, C++, Simulink, Stateflow, Scikit-learn,**Pytorch**, **TensorFlow**, Keras, pandas, git, PyQt5, pylsl, **Hydra, Mlflow**, Optuna, PostgreSQL,bleak
CAD/CAM: NX Design, PTC Creo.
Core Competencies: Control Systems, Machine Learning/ Deep Learning, **HIL**, RNN, **Bayesian Statistics**, **Linux Environment, Assistive robotics,** State Estimation**, Kalman Filters**,**Sensor Fusion**.
Personal Attributes: Can-Do attitude, Proactive, Problem solver

Education

Master of Science, University of Illinois at Chicago — Jan 2021- Jan 2023

Publication: Vakacherla, S. S., Kantharaju, P., Mevada, M., and Kim, M. (February 6, 2023). "**Single Accelerometer to Recognize Human Activities Using Neural Networks.**" ASME. *J Biomech Eng*. June 2023; 145(6): 061005. https://doi.org/10.1115/1.4056767
Relevant coursework: Dynamics of mechanical system, Mechatronics, Dynamics of control systems, Assistive Robotics, Independent Study on Battery Management Systems

Bachelor of Engineering, Maharaja Sayajirao University — Apr 2015-Apr2019

GPA: 3.93/4.0
Relevant coursework: Dynamics of Machine, Computer applications in MechanicalEngineering, Machine design

Professional Experience

Automation Engineer, Hariome Spring Industries — Oct 2019 – Oct 2020

- In a span of one year, we designed and made a spring manufacturing machine that can manufacture springs from 0.2-5 mm wires with different pitch and length in batches.
- Constructed **3d models** of Wire drawing, wire bending and wire cutting mechanisms using **Siemens Nx.**
- Developed a **feedback control algorithm** to control 2 DC servo motors and a magnetic clutch to achieve the desired output for Arduino Mega using C++.
- Designed Human Machine Interface to give inputs to the system and operate the machine

Master's Thesis – Assistance in Natural Environment — Jan 2022

Purpose of this project is to create a higher-level controller for the exoskeleton that will help the subject in using the **exoskeleton in natural environment.** System has two components to it, a) **Human Activity Recognition,** b) **Walking Speed Estimation.** Both components use data from a **single 3 axis – Accelerometer** attached to subject's sternum. Also Created a **python package** using **pip** that contains both **Activity Recognition** and **Walking Speed Estimation** combined.

Human Activity Recognition — Jan 2022 – Aug 2022

- Using Deep learning architectures, we classified standing, squatting, walking, walking on gradient and running with **state-of-the-art** accuracy of **96%.**
- Developed 4 different architectures, CNN, CNN – LSTM - CNN, LSTM – CNN and LSTM with a SoftMax classifier at the tail using **PyTorch**.

- Trained the networks using PyTorch alongside **Hydra and MLFlow** and found two best performing architectures for further hyperparameter tuning using **Optuna**.
- Produced models with 99 % training accuracy and 97 % testing accuracy along with **96 %** online testing accuracy on two subjects.

WAKING SPEED ESTIMATION JAN 2022

Developed two different approaches of walking speed estimation using a) Model based and b) Data Based.

MODEL BASED

- Created a model based on **inverted pendulum walking model** to calculate the walking speed using **Center of Mass trajectory** with state-of-the-art MSE of **0.00357 m/s**.
- **Invented Trend Filter** and merged it with 1D **Kalman Filter** to filter the process variance and reduce the standard deviation of estimation by **74%.**

DATA BASED

- Developed a **Deep learning** architecture to estimate the walking speed using Height Normalized data from Single 3 axis Accelerometer with **0.0007 m/s MSE**.
- Created 3 different architectures CNN, CNN – LSTM – CNN and Transformer Encoder – CNN for regression and trained with **KFold** Cross validation using **PyTorch** And tracked using Weights and Biases.
- Designed an **Iterative Learning** algorithm to tune parameters of model-based approach using estimation of Data Based approach online
- Created a **UI** to **combine both** Activity Recognition and Walking Speed estimation using Pyqt5

GRADUATE RESEARCHER: UNIVERSITY OF ILLINOIS AT CHICAGO AUG 2021

BECKHOFF TWINCAT DATA ACQUISITION SYSTEM (DAQ) AUG 2022 – JAN 2023

- Designed DAQ with Beckhoff TwinCAT digital and analog modules for exoskeleton sensors.
- Interfaced sensor signals from TwinCAT to Simulink controller.
- Created UDP communication script with C++ between TwinCAT and walking speed estimation system.

EXOSKELETON ASSISTANCE OPTIMIZATION NOV 2021

- Objective of the project is to optimize the assistance parameters of an **ankle exoskeleton** to individual subjects and to different walking speeds
- Conducted **Gaussian process** to estimate the distribution of metabolic cost against walking speed and assistance parameters.
- Created a one parameter online **Bayesian Optimization** algorithm to perform Human in Loop optimization to find best parameters of the ankle exoskeleton at different walking speeds using **MATLAB**.
- Reduced the metabolic cost of walking at different speeds by average 10.5% by performing Human in Loop optimization for 4 subjects.
- Designed a python program to change assistance parameter of exoskeleton depending on the walking speed estimation of our system using **UDP** communication between two computers.

VICON (MOTION CAPTURE SYSTEM) DATA STREAMING SOFTWARE JAN 2022

- Developed a **python pip package** and c++ program to live stream the data of motion capture system into the lab streaming layer to make a centralized data collection system for the experiments.
- Designed the algorithm to stream Labeled Markers, Unlabeled Markers and coupled Devices' data with minimum latency.
- Devised the program such that it's robust against occlusion of the markers and dropping of markers from frames along with the stream containing the occlusion data.
- This helped reducing the error in data collection and made the experiments complete faster by 20%.
- Developing a program that can save all the collected data with different formats into one PostgreSQL server.
- Designed a software to start / stop and record all the pylsl streams using threading, multiprocessing, PyQt5 and lab recorder.

PROJECTS

MULTIBOT MATERIAL HANDLING SYSTEM — DEC 2018- APR 2019

- Built an autonomous system comprising of 3 BOTs/AGVs that can work in synchronization with each other to carry an object either alone or with multiple BOTs depending on the load of the object.
- Each BOT had four stepper motors with differential drive for locomotion and one DC motor for handling the material and uses nRF modules to communicate with each other, all controlled by a microcontroller (Arduino Uno).
- Designed the control algorithm to actuate the BOTs along their respective pathways using C++.
- Used Master/slave configuration between BOTs and the central controller that commands the BOTs.

VEHICLE DETECTION USING MASK RCNN — AUG 2020- NOV 2020

- Trained a RCNN to detect street objects like pedestrians, traffic signals and street signs, and vehicles in an image.
- The RCNN was comprised of Region Proposal Network which worked on selective search algorithm, a feature extraction CNN, SVM to classify the object, a bounding box regressor and a masking network.
- Instead of training the whole network on a dataset, I used transfer learning to speed up training.
- The network was trained and tested on Streetscenes database of Center for Biological & Computational Learning with k=5 fold cross validation with training accuracy of 76.23%.

MULTIBODY SYSTEM SIMULATION — JAN 2021- APR 2021

- Developed a generic MATLAB program that can kinematically simulate some simple planar kinematic chains with any number of bodies with joints like ground, prismatic and revolute.
- Used symbolic expressions to make it more robust and generic so that it can be used for any kinematic chain.
- Utilized constraint kinematics and Newton-Raphson method to solve the equations.

CERTIFICATION

- **IPS for researchers – CITI Program**
- Machine Learning by Stanford online
- MATLAB Fundamentals
- Structuring Machine Learning project
- HSP, Biomedical Research Investigator and Key Personnel- CITI Program
- Neural Network and Deep learning
- Python essential training
- Simulink Onramp

EXTRA-CURRICULAR ACTIVITIES

- Presented unique catalytic converter design to reduce the emissions at MG innovation Challenge.
- Won 2 tournaments back-to-back in 2 years as a **Team Captain** for the college cricket team.
- Represented State in **Interstate** tournament of Cricket at Under 16 level.